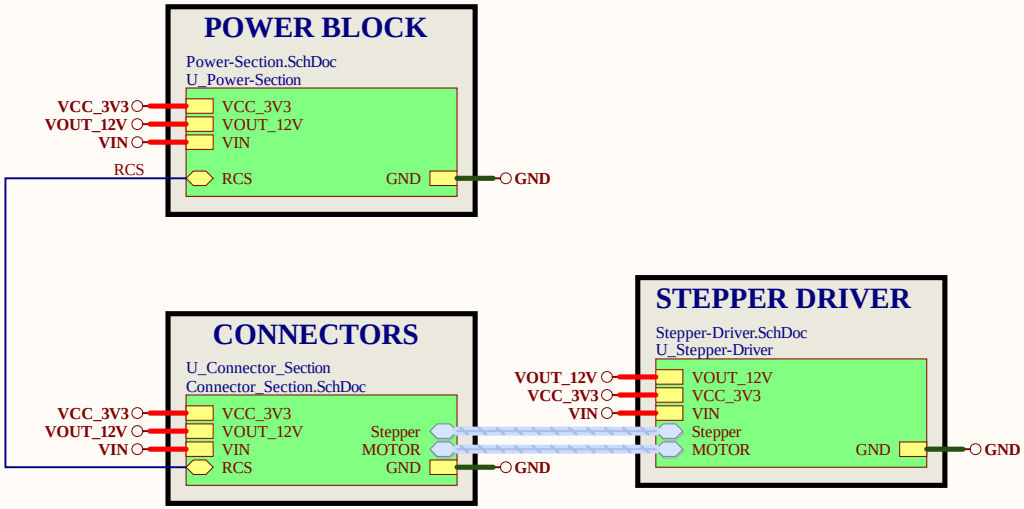
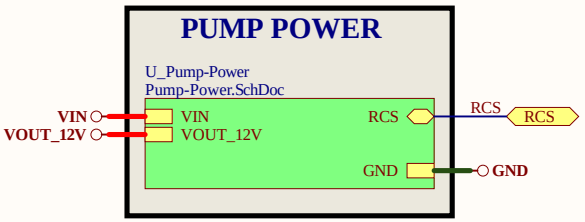
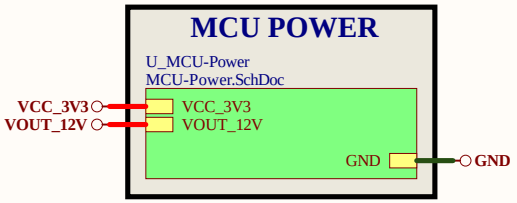
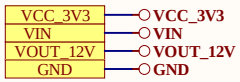
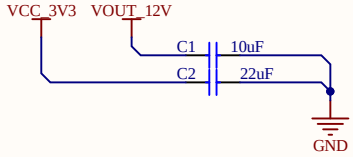

POWER BLOCK
Power-Section.SchDoc
U_Power-Section
VCC_3V3
VOUT_12V
VIN
RCS
VCC_3V3
VOUT_12V
VIN
RCS
GND
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VCC_3V3
VOUT_12V
VIN
RCS
VCC_3V3
VOUT_12V
VIN
RCS
Stepper
MOTOR
GND
GND
STEPPER DRIVER
Stepper-Driver.SchDoc
U_Stepper-Driver
VOUT_12V
VCC_3V3
VIN
VOUT_12V
VCC_3V3
VIN
Stepper
MOTOR
GND
GND


PUMP POWER
U_Pump-Power
Pump-Power.SchDoc
VIN
VOUT_12V
VIN
VOUT_12V
RCS
GND
RCS
RCS
GND
GND

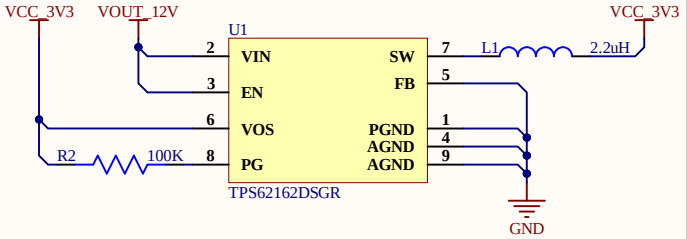

MCU POWER
U_MCU-Power
MCU-Power.SchDoc
VCC_3V3
VOUT_12V
VCC_3V3
VOUT_12V
GND
GND


VCC_3V3
VIN
VOUT_12V
GND
VCC_3V3
VIN
VOUT_12V
GND

DECOUPLING CAPACITORS

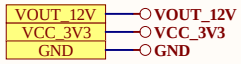


3V3 BUCK REGULATOR

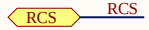


Fixed regulator.
VOUT= 3V3/1A

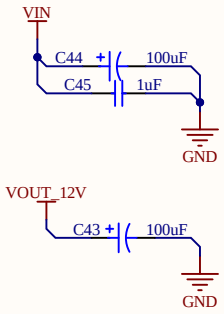
POWER SIGNALS



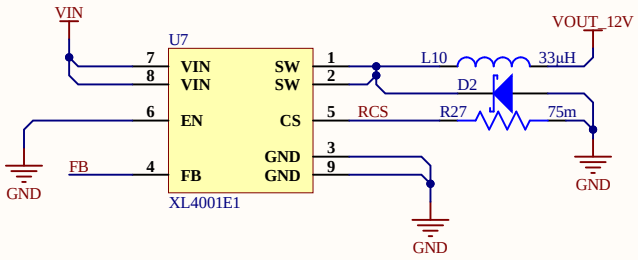
CURRENT SENSE INPUT



DECOUPLING CAPACITORS



12V BUCK REGULATOR

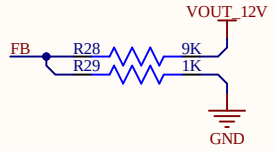


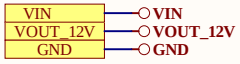
VIN=36V
VOUT=12V, where VOUT= 1.235(1+R2/R1)*
For 12V output, R2=9K and R1=1K
(Datasheet pg.3)

For IOUT=2A, RCS=75mA, (RCS=0.155/IOUT)

Reference: XL4001 Datasheet

FEEDBACK NETWORK

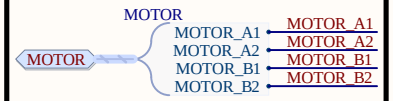




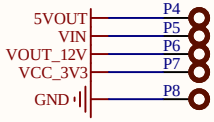
CONTROL SIGNALS



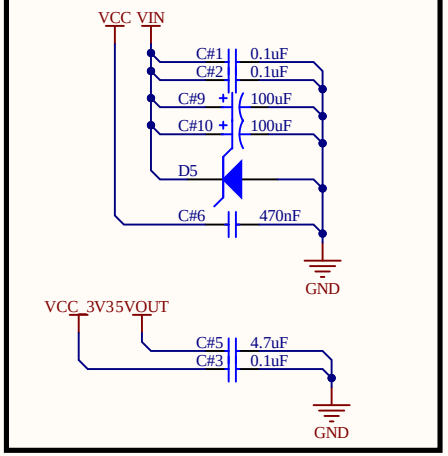
MOTOR OUT



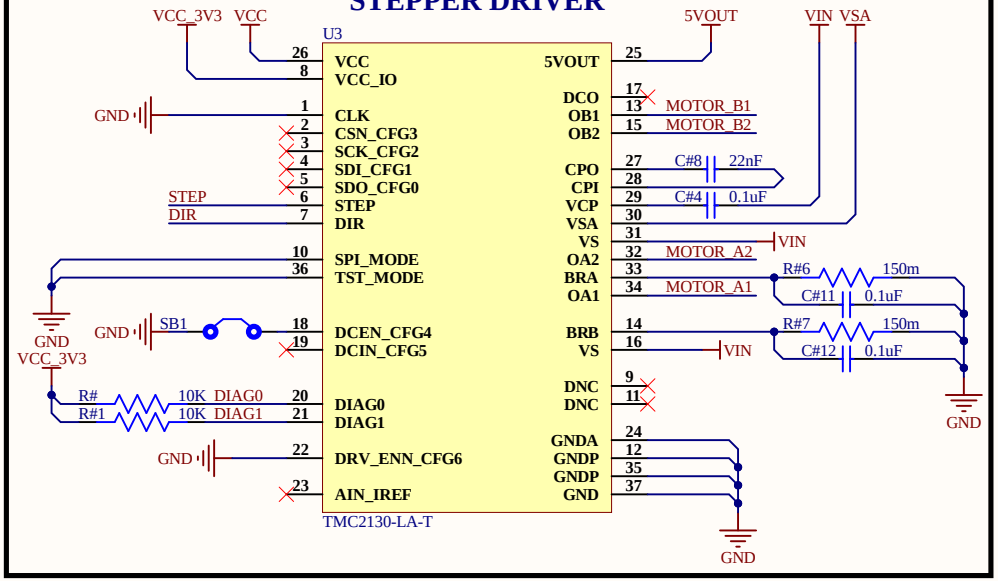
TEST POINTS



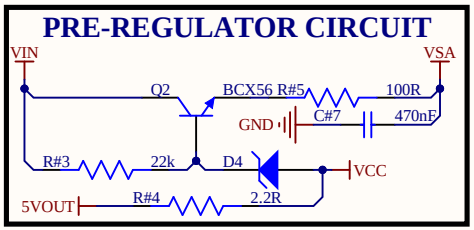
DECOUPLING CAPACITORS



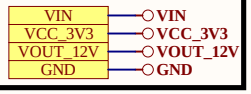
STEPPER DRIVER

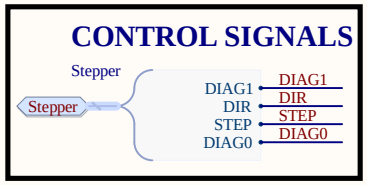


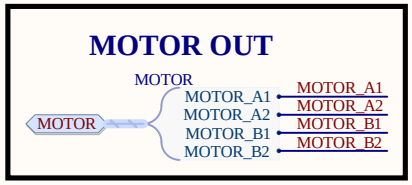
PRE-REGULATOR CIRCUIT

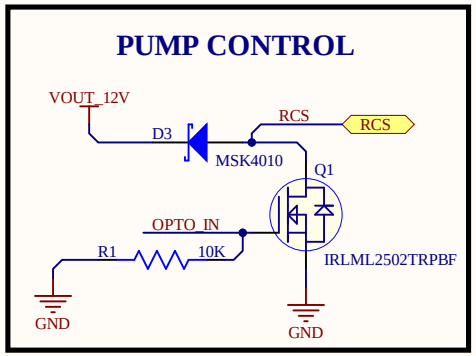


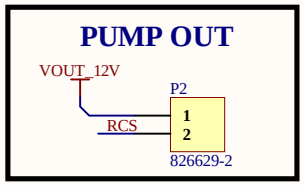
POWER SIGNALS

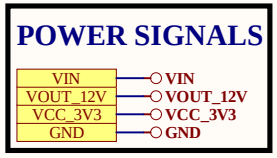


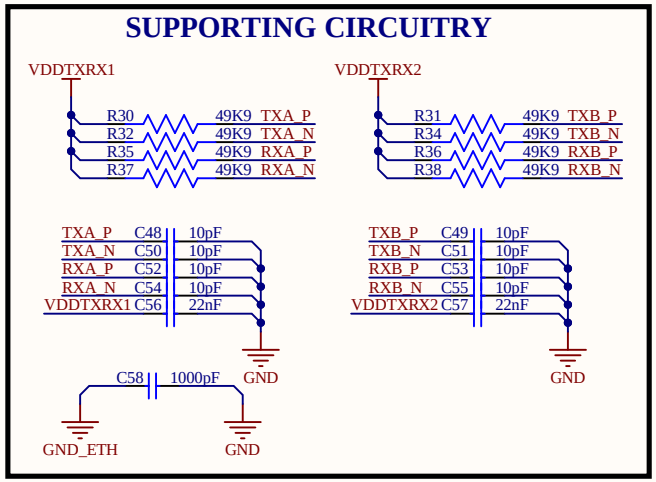

CONTROL SIGNALS
Stepper
Stepper
DIAG1
DIR
STEP
DIAG0
DIAG1
DIR
STEP
DIAG0

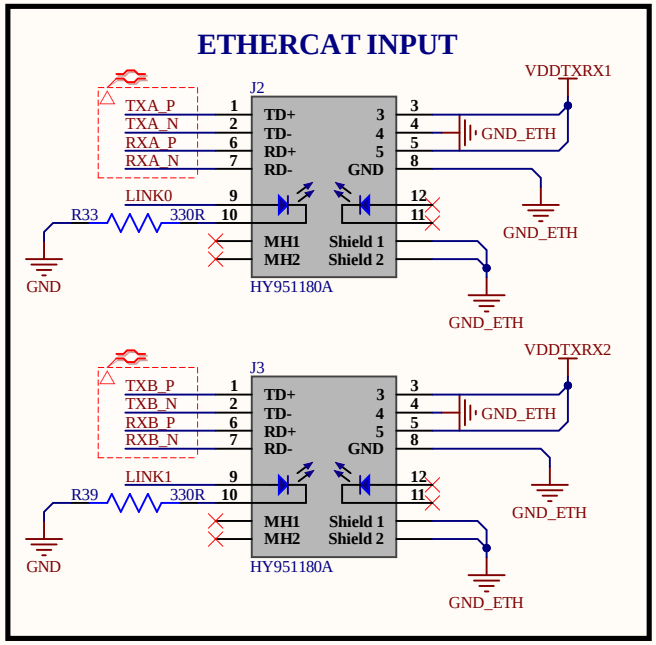

MOTOR OUT
MOTOR
MOTOR
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2

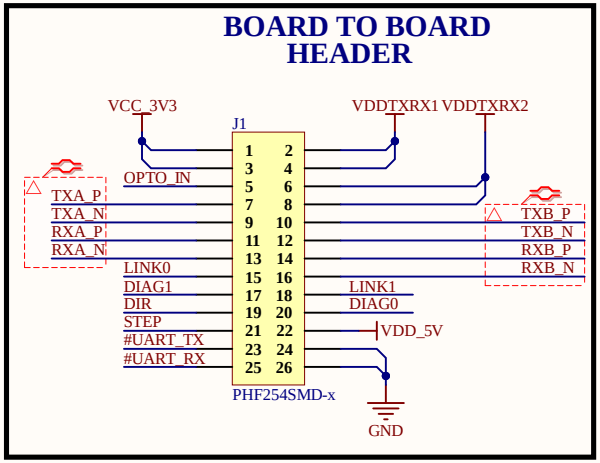

PUMP CONTROL
VOUT_12V
D3
MSK4010
RCS
RCS
Q1
OPTO_IN
R1
10K
IRLML2502TRPBF
GND
GND

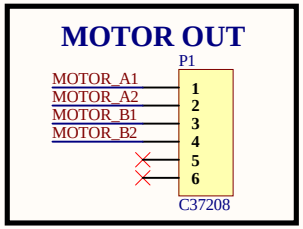

PUMP OUT
VOUT_12V
RCS
P2
1
2
826629-2

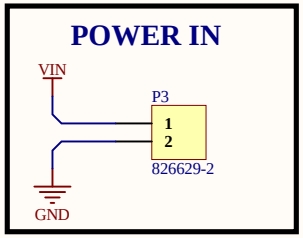

POWER SIGNALS
VIN
VOUT_12V
VCC_3V3
GND
VIN
VOUT_12V
VCC_3V3
GND

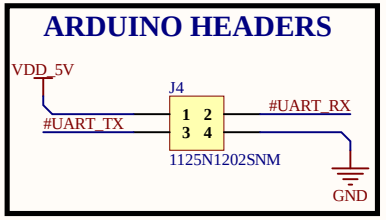

SUPPORTING CIRCUITRY
VDDTXRX1
VDDTXRX2
R30 49K9 TXA_P
R32 49K9 TXA_N
R35 49K9 RXA_P
R37 49K9 RXA_N
R31 49K9 TXB_P
R34 49K9 TXB_N
R36 49K9 RXB_P
R38 49K9 RXB_N
TXA_P C48 10pF
TXA_N C50 10pF
RXA_P C52 10pF
RXA_N C54 10pF
VDDTXRX1 C56 22nF
TXB_P C49 10pF
TXB_N C51 10pF
RXB_P C53 10pF
RXB_N C55 10pF
VDDTXRX2 C57 22nF
GND
GND
C58 1000pF
GND_ETH
GND


ETHERCAT INPUT
VDDTXRX1
J2
TXA_P 1
TXA_N 2
RXA_P 6
RXA_N 7
TD+ 3
TD- 4
RD+ 5
RD- GND 8
3 4 5 8
GND_ETH
LINK0 9
R33 330R 10
12 11
MH1 Shield 1
MH2 Shield 2
HY951180A
GND
GND_ETH
GND_ETH
VDDTXRX2
J3
TXB_P 1
TXB_N 2
RXB_P 6
RXB_N 7
TD+ 3
TD- 4
RD+ 5
RD- GND 8
3 4 5 8
GND_ETH
LINK1 9
R39 330R 10
12 11
MH1 Shield 1
MH2 Shield 2
HY951180A
GND
GND_ETH
GND_ETH


BOARD TO BOARD HEADER
VCC_3V3
VDDTXRX1 VDDTXRX2
J1
OPTO_IN
TXA_P
TXA_N
RXA_P
RXA_N
TXB_P
TXB_N
RXB_P
RXB_N
LINK0
DIAG1
DIR
STEP
#UART_TX
#UART_RX
LINK1
DIAG0
VDD_5V
PHF254SMD-x
GND


MOTOR OUT
MOTOR_A1
MOTOR_A2
MOTOR_B1
MOTOR_B2
P1
1
2
3
4
5
6
C37208


POWER IN
VIN
P3
1
2
826629-2
GND


ARDUINO HEADERS
VDD_5V
J4
#UART_TX
#UART_RX
1 2
3 4
1125N1202SNM
GND